

# XAMPP Hosting Guide

What XAMPP is, why it is the right platform for the US Tape Time Tracking Application, and how to install, secure and run it.

|                       |                                                                          |
|-----------------------|--------------------------------------------------------------------------|
| <b>Audience</b>       | US Tape IT staff and the developers who maintain the app after handoff   |
| <b>Applies to</b>     | Branch <b>main</b> of the timeclock repository, September 2026           |
| <b>Tested with</b>    | XAMPP 8.2.4 (PHP 8.2.4, MariaDB 10.4.28) and PHP 8.5 / MySQL 8.0         |
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| <b>Companion docs</b> | README.md, Git & Version Control Guide (PDF), ISSUES_AND_IMPROVEMENTS.md |

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# 1. Summary

The US Tape Time Tracking Application is a set of plain PHP pages backed by a MySQL-compatible database. To run, it needs exactly three things: a **web server**, the **PHP** interpreter, and a **database**. XAMPP is a free package that installs all three, plus a browser-based database tool, in one step on a single Windows PC.

**Recommendation:** host the application with XAMPP on one always-on PC inside the US Tape network, harden the default settings (Section 7), and schedule nightly database backups (Section 8). This meets the intranet-only requirement confirmed with US Tape IT, needs no code changes, and can be run by IT staff without Linux or web-server administration experience.

## Read this before go-live

- XAMPP's defaults are built for convenience, not security: the database root account has no password. Section 7 is **mandatory**, not optional.
- The application itself still has open security items (plain-text passwords, unprotected legacy pages). They are listed in ISSUES\_AND\_IMPROVEMENTS.md and must be fixed before real payroll use, whatever the hosting platform.

# 2. What XAMPP is

XAMPP is a free, open-source bundle from the Apache Friends project. It packages the most common open-source web stack into one installer with a graphical Control Panel. The name lists what is inside:

| Letter | Component                  | What it does                                        | Used by this app?                               |
|--------|----------------------------|-----------------------------------------------------|-------------------------------------------------|
| X      | Cross-platform             | Same package on Windows, macOS and Linux            | Yes: developed on macOS, deployed on Windows    |
| A      | Apache HTTP Server         | Receives browser requests and returns web pages     | <b>Yes:</b> serves every page                   |
| M      | MariaDB (MySQL fork)       | Relational database that stores the data            | <b>Yes:</b> stores employees, shifts, audit log |
| P      | PHP                        | Server-side language that builds each page          | <b>Yes:</b> the whole app is PHP                |
| P      | Perl                       | Older scripting language                            | No (leave it unused)                            |
|        | phpMyAdmin (bonus)         | Web interface for browsing and editing the database | <b>Yes:</b> setup, fixes, backups               |
|        | FileZilla, Mercury, Tomcat | FTP, mail and Java servers (Windows build)          | No: do not install                              |

Without XAMPP, each of these would be downloaded, installed and configured separately, and each would need to be pointed at the others. XAMPP ships them already wired together, so Apache runs PHP, PHP can reach MariaDB, and everything starts from one Control Panel.

## MariaDB vs. MySQL

XAMPP ships **MariaDB**, a community fork of MySQL made by MySQL's original developers. For this application the two are interchangeable: the same SQL, the same `mysql: PDO` driver, and the same tools. The team developed against MySQL 8 and the CI pipeline tests against MySQL 8. The JSON columns in `AUDIT_LOG` are supported by MariaDB 10.2 and later.

## 3. How the application runs on XAMPP

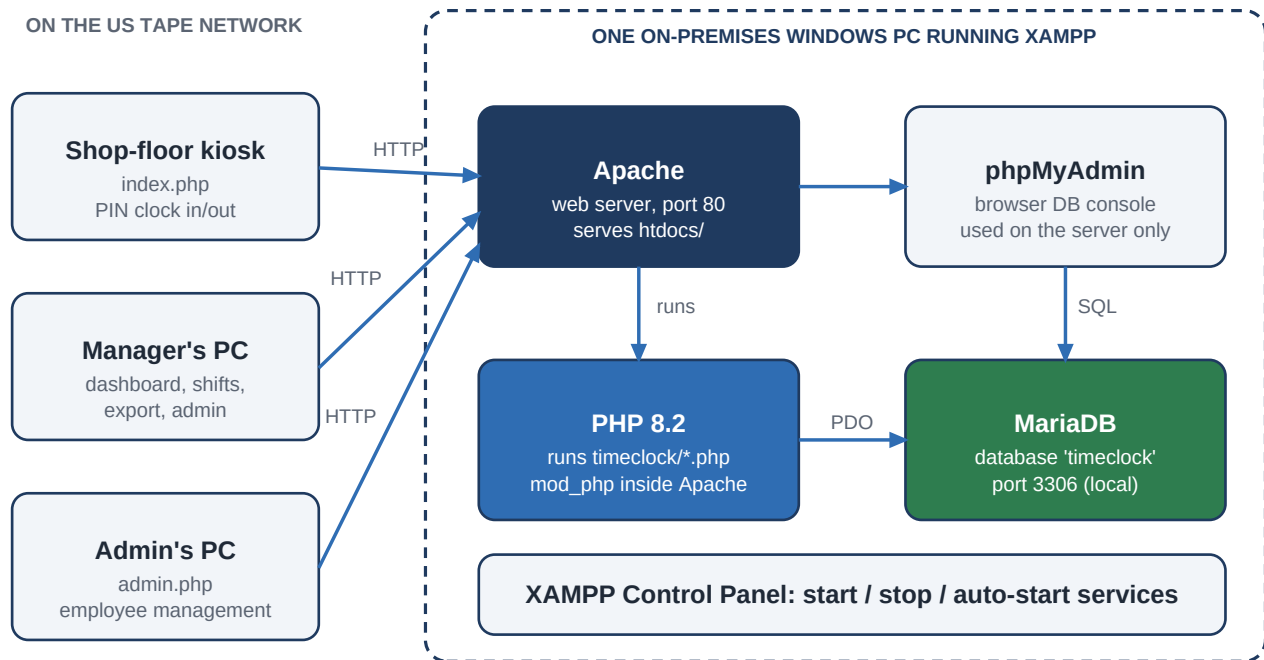


Figure 1: Deployment layout. Every device talks to one PC; the database is never exposed to the network.

Every page in the repository is a separate PHP file. When a browser asks for `http://SERVER/timeclock/index.php`, Apache hands the file to PHP. PHP runs it, queries MariaDB through PDO (configured in `config.php`), and returns plain HTML. There is no build step, no Node.js, no framework and no Composer packages on `main`, so the files in the repository are exactly the files that run.

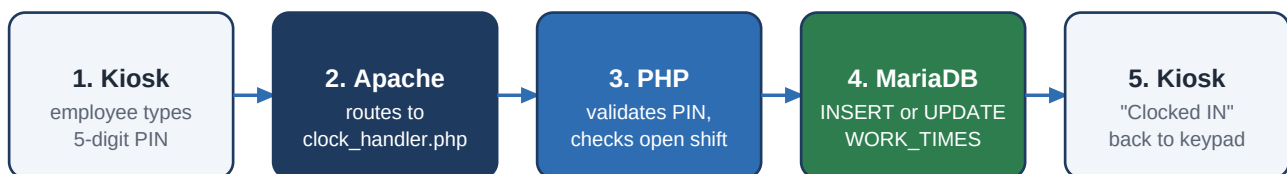


Figure 2: What happens when an employee punches in.

## Where things live in XAMPP

| Item              | Windows (production)           | macOS (developer machines)                    |
|-------------------|--------------------------------|-----------------------------------------------|
| Install folder    | C:\xampp                       | /Applications/XAMPP                           |
| Web root (htdocs) | C:\xampp\htdocs                | /Applications/XAMPP/htdocs                    |
| This app          | C:\xampp\htdocs\timeclock      | /Applications/XAMPP/htdocs/timeclock          |
| App URL           | http://SERVER-NAME/timeclock/  | http://localhost/timeclock/                   |
| phpMyAdmin        | http://localhost/phpmyadmin    | http://localhost/phpmyadmin                   |
| PHP settings      | C:\xampp\php\php.ini           | /Applications/XAMPP/xamppfiles/etc/php.ini    |
| Database settings | C:\xampp\mysql\bin\my.ini      | /Applications/XAMPP/xamppfiles/etc/my.cnf     |
| Apache error log  | C:\xampp\apache\logs\error.log | /Applications/XAMPP/xamppfiles/logs/error_log |
| Control Panel     | C:\xampp\xampp-control.exe     | manager-osx.app                               |

## 4. Why XAMPP is the best fit for this project

The platform was chosen against this project's actual constraints: the app must stay inside the US Tape network, the app is plain PHP + MySQL, the people running it day to day are IT generalists rather than web-server specialists, and development is handed from one student team to the next.

### 4.1 It matches the application exactly, with zero code changes

The app needs Apache (or any PHP-capable web server), PHP and a MySQL-compatible database, and XAMPP provides exactly those. Before writing this guide, every PHP file on main was syntax-checked with XAMPP's own PHP 8.2.4 and all of them passed. Moving from a developer laptop to XAMPP means copying the folder and creating `config.php`.

### 4.2 It satisfies the intranet-only requirement

US Tape IT confirmed that this system must stay on the internal network (confirmed 2026-04-14). XAMPP runs entirely on one on-premises machine. Employee names, PINs and hours never leave the building, nothing depends on an internet connection, and there is no monthly hosting bill. The app loads no CDNs or external scripts, so the kiosks keep working even if the internet is down.

### 4.3 Anyone in IT can operate it

Everything is managed from one Control Panel window: green means running, and Start/Stop buttons and log links sit next to each service. Apache and MariaDB can be installed as Windows services with one checkbox so they start automatically after a reboot. No command line, Linux knowledge or web-server configuration files are needed for normal operation.

## 4.4 phpMyAdmin is built in

Some jobs can't be done in the app yet, such as creating the first admin, fixing a bad row or running a schema migration. phpMyAdmin lets an authorized person do these in a browser, and it can also export a full backup in two clicks.

## 4.5 It survives team turnover

XAMPP is the stack most PHP courses and tutorials use, so a new student team or contractor will usually already know it. Setting up a developer laptop takes the same steps as setting up the server, so "works on my machine" problems are rare.

## 4.6 It's free and has a clear upgrade path

XAMPP has no license fees. It is also the *standard* Apache/PHP/MySQL stack, so it doesn't lock US Tape in. If the app grows, it can move to a hardened Windows or Linux server by copying the `timeclock` folder and restoring a database dump, with no code changes.

## 4.7 Alternatives considered

| Option                                                             | Strengths                                                                      | Why it wasn't chosen (for now)                                                                                     |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>XAMPP on an on-prem PC</b><br>(recommended)                     | One installer, GUI control, phpMyAdmin, free, intranet-only, exact stack match | Insecure defaults. Fixed by the Section 7 checklist                                                                |
| PHP built-in server + standalone MySQL (current dev setup)         | Fastest way to develop; nothing extra to install                               | Handles one request at a time, isn't a service, doesn't survive a reboot. The PHP docs say it's not for production |
| Separately installed Apache + PHP + MySQL (or a Linux LAMP server) | Most production-grade; full control                                            | Needs server-admin skills to install, configure and patch each part. Good upgrade target later                     |
| IIS + PHP on Windows Server                                        | Fits a Microsoft-standard IT shop                                              | FastCGI, URL and PHP setup by hand; MySQL still separate; harder for student developers to copy locally            |
| Docker containers                                                  | Reproducible environments                                                      | Adds Docker Desktop, virtualization and a new skill set for IT; too much for a single small app                    |
| Cloud or shared web hosting                                        | No hardware to manage                                                          | Breaks the intranet-only requirement: employee data leaves the building; recurring cost                            |

## 4.8 Honest trade-offs

- **XAMPP is designed for development first.** Apache Friends says so themselves. It is suitable here because the app is internal, low-traffic and hardened per Section 7. It would *not* be the right choice for a public, internet-facing site.

- **Upgrades are manual.** A new XAMPP version means installing it next to the old one and moving the database across (Section 8.4). Plan to check for PHP security releases a few times a year.
- **Single machine.** If the PC dies, the clock goes down until it's restored from backup. For a small shop that is acceptable *if* nightly backups are copied to another machine.

## 5. Installing XAMPP (Windows server)

Before starting, ask IT for: an always-on Windows PC on the internal network, a **static IP address or DNS name** for it (for example `timeclock.ustape.local`), and local administrator rights for the install.

1. Download **XAMPP for Windows, PHP 8.2 or newer** from <https://www.apachefriends.org>.
2. Run the installer **as Administrator**. On the components screen keep **Apache, MySQL, PHP, phpMyAdmin**. Untick FileZilla, Mercury, Tomcat, Perl, Webalizer and Fake Sendmail.
3. Install to `C:\xampp`. Avoid `C:\Program Files`, where Windows permission controls (UAC) cause write errors.
4. When Windows Firewall prompts for Apache, allow **Private** networks only.
5. Open the **XAMPP Control Panel as Administrator**. Tick the **Svc** box next to Apache and MySQL to install both as Windows services, so they start automatically at boot.
6. Click **Start** for Apache and MySQL. Both rows should turn green.
7. Browse to `http://localhost/` on the server. The XAMPP welcome page confirms Apache and PHP are working.

### Developer laptops (macOS)

Install the macOS build to `/Applications/XAMPP` and use `manager-osx.app` to start services. If another MySQL already uses port 3306 (for example Homebrew MySQL), XAMPP's MariaDB won't start. Either stop the other one or change XAMPP's port. On the original dev Mac, XAMPP MariaDB was moved to **3307**, so `config.php` must use that port there.

## 6. Deploying the application

### 6.1 Put the code in htdocs

Preferred: install Git for Windows and clone, so later updates are a single `git pull`. See the Git guide.

```
cd C:\xampp\htdocs
git clone git@github.com:Saul2525/US-Tape---Time-Tracking-Application.git timeclock
```

Or copy the `timeclock` folder into `C:\xampp\htdocs\`. Either way, `index.php` must end up at `C:\xampp\htdocs\timeclock\index.php`.

### 6.2 Create and load the database

1. Open `http://localhost/phpmyadmin` **on the server**.
2. **Databases** tab → name `timeclock`, collation `utf8mb4_general_ci` → Create.
3. Select `timeclock` → **Import** → choose `db.sql` from the repository → Import.

4. Open the **SQL** tab and run the migration below. `db.sql` is missing these changes, and without them no manager can log in.

```
ALTER TABLE EMPLOYEES ADD COLUMN password VARCHAR(255) NULL;  
ALTER TABLE EMPLOYEES MODIFY pin CHAR(5) NOT NULL;  
ALTER TABLE EMPLOYEES ADD UNIQUE (pin);
```

## 6.3 Create a dedicated database user

The app should not connect as root. In the SQL tab:

```
CREATE USER 'timeclock_app'@'localhost' IDENTIFIED BY 'a-long-random-password';  
GRANT SELECT, INSERT, UPDATE, DELETE ON timeclock.* TO 'timeclock_app'@'localhost';  
FLUSH PRIVILEGES;
```

## 6.4 Create config.php

`config.php` is deliberately left out of Git so passwords are never committed. Create `C:\xampp\htdocs\timeclock\config.php`:

```
<?php  
$host = "127.0.0.1";  
$port = 3306; // must match port= in C:\xampp\mysql\bin\my.ini  
$db = "timeclock";  
$user = "timeclock_app";  
$pass = "a-long-random-password";  
  
try {  
    $pdo = new PDO("mysql:host=$host;port=$port;dbname=$db;charset=utf8mb4", $user, $pass);  
    $pdo->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);  
} catch (PDOException $e) {  
    error_log($e->getMessage());  
    die("The time clock is temporarily unavailable. Please tell a manager.");  
}
```

## 6.5 Set the PHP timezone

In `C:\xampp\php\php.ini` set `date.timezone = America/New_York`, then restart Apache from the Control Panel. The pages' date filters use this timezone. Punches are stored in UTC, which is a known display issue tracked in `ISSUES_AND_IMPROVEMENTS.md`.

## 6.6 Create the first admin, then test

```
INSERT INTO EMPLOYEES (first_name, last_name, email, pin, password, role_id, is_active)  
VALUES ('First', 'Admin', 'admin@ustape.local', '10000', 'ChangeMe!123', 3, 1);
```

1. From **another PC** on the network, open `http://SERVER-NAME/timeclock/`. The PIN keypad should appear.



2. Click **Manager Sign In** and log in as the admin. Open **Employee Management** and create a test employee, noting the PIN.
3. Punch that PIN in and out on the kiosk page, then confirm the shift appears under **View Shifts** and exports with **Export to Excel**.
4. Deactivate the test employee and change the admin's password.

## 6.7 Set up the kiosks

On each shop-floor device, set the browser home page to `http://SERVER-NAME/timeclock/` and run it full-screen (Chrome and Edge both have a kiosk mode). Bookmarking by DNS name instead of IP means kiosks keep working if the server's IP changes.

## 7. Security hardening checklist (mandatory)

XAMPP ships with open defaults so developers can get started quickly. Do all of the following before real employees use the system.

| #  | Action                            | How                                                                                                                                                                                                |
|----|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Set a MariaDB root password       | phpMyAdmin → User accounts → root@localhost → Edit privileges → Change password. Then set <code>auth_type = 'cookie'</code> in <code>C:\xampp\phpMyAdmin\config.inc.php</code>                     |
| 2  | App uses its own DB user          | Section 6.3. Never connect as root from config.php                                                                                                                                                 |
| 3  | Keep the database off the network | In <code>my.ini</code> under <code>[mysqld]</code> add <code>bind-address=127.0.0.1</code> . Don't open port 3306 in the firewall                                                                  |
| 4  | Keep phpMyAdmin local-only        | In <code>C:\xampp\apache\conf\extra\httpd-xampp.conf</code> , make sure the phpMyAdmin block says <code>Require local</code>                                                                       |
| 5  | Remove the XAMPP demo pages       | Move <code>htdocs\dashboard</code> , <code>img</code> , <code>webalizer</code> etc. out of <code>htdocs</code> ; replace <code>htdocs\index.php</code> with a redirect to <code>/timeclock/</code> |
| 6  | Hide errors from users            | <code>php.ini</code> : <code>display_errors = Off</code> , <code>log_errors = On</code> , <code>expose_php = Off</code>                                                                            |
| 7  | Firewall port 80 to the LAN       | Windows Defender Firewall: allow inbound TCP 80 (and 443 if HTTPS) from the internal subnet only                                                                                                   |
| 8  | Remove the legacy pages           | Delete or lock <code>create_user.php</code> , <code>dashboard.php</code> , <code>profile.php</code> . They have no login check                                                                     |
| 9  | Enable HTTPS (recommended)        | XAMPP includes <code>mod_ssl</code> . Ask IT for a certificate from the internal CA. HTTPS is also required if clock-in geolocation is added later                                                 |
| 10 | Fix app-level P0 issues           | Hash manager passwords, add permission check to <code>update_shift.php</code> , rate-limit PIN attempts. See <code>ISSUES_AND_IMPROVEMENTS.md</code>                                               |

## 8. Day-to-day operations

### 8.1 Starting and stopping

With the services installed (Section 5, step 5), Apache and MariaDB start on boot by themselves. To restart after a config change, use the Control Panel's Stop/Start buttons, or **services.msc** → *Apache2.4* / *mysql*.

### 8.2 Nightly backups

Save this as `C:\timeclock-backup\backup.bat` and schedule it daily with Windows **Task Scheduler**. Use a backup-only DB user with SELECT and LOCK TABLES rights, and copy the output folder to a network share or another machine.

```
@echo off
set STAMP=%date:~-4%-%date:~4,2%-%date:~7,2%
"C:\xampp\mysql\bin\mysqldump.exe" -u timeclock_backup -pBACKUP_PASSWORD ^
--single-transaction --routines timeclock > "C:\timeclock-backup\timeclock_%STAMP%.sql"
forfiles /p "C:\timeclock-backup" /m *.sql /d -60 /c "cmd /c del @path"
```

Restore (tested at least once before go-live): `C:\xampp\mysql\bin\mysql.exe -u root -p timeclock < timeclock_YYYY-MM-DD.sql`

### 8.3 Forgotten clock-out check

`cron_anomaly.php` lists shifts open longer than 10 hours. It prints only to the console for now. To keep a log, schedule this in Task Scheduler:

```
C:\xampp\php\php.exe C:\xampp\htdocs\timeclock\cron_anomaly.php >> C:\timeclock-backup\anomalies.log
```

### 8.4 Updating the application

1. Run the backup script manually.
2. `cd C:\xampp\htdocs\timeclock` then `git pull`.
3. Run any new SQL migration the release notes mention, in phpMyAdmin.
4. Smoke-test one punch and one manager login.

### 8.5 Upgrading XAMPP itself

Install the new XAMPP version to a new folder (for example `C:\xampp-new`) while the old one is stopped. Copy the `timeclock` folder, restore the latest backup into the new MariaDB, re-apply Section 7, and switch the services over. Keep the old folder until the new one has run for a week.

## 9. Troubleshooting

| Symptom                                         | Likely cause                                                  | Fix                                                                                                                            |
|-------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Apache won't start (red)                        | Port 80 is in use by IIS, Skype or another web server         | <code>netstat -ano   findstr :80</code> to find it; stop that service or change <code>Listen</code> in <code>httpd.conf</code> |
| MySQL won't start                               | Port 3306 is in use by another MySQL                          | Stop the other MySQL, or change <code>port=</code> in <code>my.ini</code> <i>and</i> in <code>config.php</code>                |
| "MySQL shutdown unexpectedly" after a power cut | Corrupted data files                                          | Read <code>mysql\data\mysql_error.log</code> . Restore from the latest backup. Don't delete the data folder by guesswork       |
| "DB Connection failed ... [2002]"               | MariaDB stopped or wrong port                                 | Start MySQL; check the port in <code>config.php</code>                                                                         |
| "... [1049] Unknown database"                   | Database not created                                          | Section 6.2                                                                                                                    |
| "... [1045] Access denied"                      | Wrong DB username/password in <code>config.php</code>         | Match <code>config.php</code> to the user from Section 6.3                                                                     |
| "Unknown column 'password'"                     | <code>db.sql</code> migration not run                         | Run the ALTER TABLE block in Section 6.2                                                                                       |
| Manager login says "Access denied."             | Account's <code>role_id</code> isn't 2 (Manager) or 3 (Admin) | Fix the role in Employee Management or phpMyAdmin                                                                              |
| Blank white page                                | PHP error with <code>display_errors</code> off                | Check the Apache error log (Section 3 table)                                                                                   |
| Shift times look 4–5 hours ahead                | Known issue: punches are stored in UTC and shown raw          | Tracked in <code>ISSUES_AND_IMPROVEMENTS.md</code> (Timezone mismatch)                                                         |
| Kiosks can't reach the page                     | Firewall, or the server's IP changed                          | Allow TCP 80 on the private profile; use a DNS name or reserved IP                                                             |

## 10. Reference: versions and environments

Minimum supported: PHP 8.2 with the `pdo_mysql` extension (enabled by default in XAMPP), and MariaDB 10.4+ or MySQL 8.0+.

| Environment                  | Web server            | PHP   | Database                   | Notes                                  |
|------------------------------|-----------------------|-------|----------------------------|----------------------------------------|
| Production (recommended)     | XAMPP Apache          | 8.2+  | XAMPP MariaDB              | Windows PC on the US Tape LAN          |
| Original dev Mac (XAMPP)     | Apache (XAMPP 8.2.4)  | 8.2.4 | MariaDB 10.4.28, port 3307 | Port moved to avoid the Homebrew MySQL |
| Original dev Mac (daily dev) | php -S 127.0.0.1:8000 | 8.5.2 | Homebrew MySQL, port 3306  | Holds the current dev/test data        |
| GitHub Actions CI            | php -S                | 8.5   | MySQL 8.0 container        | Branch ci-testing only                 |